

$$\begin{aligned}
\mathbf{h} &= \text{Dropout}(\text{QuickGELU}(\mathbf{W}_{\text{down}}\mathbf{x} + \mathbf{b}_{\text{down}})), \\
\text{Adapter}(\mathbf{x}) &= \mathbf{W}_{\text{up}}\mathbf{h} + \mathbf{b}_{\text{up}}, \\
\mathbf{x}' &= \mathbf{x} + \text{Adapter}(\mathbf{x}).
\end{aligned} \tag{1}$$

$$\begin{aligned}
\mathcal{P}_c^{\text{base}} &= \text{Concat}(\mathcal{P}_{\text{pre}}, [\mathbf{v}_{c,0}, \mathbf{v}_{c,1}, \mathbf{v}_{c,2}, \mathbf{v}_{c,3}], \mathcal{P}_{\text{suf}}(c)), \\
\mathcal{P}_c^{\text{geo}} &= \text{Concat}(\mathcal{P}_{\text{pre}}, [\mathbf{v}_{c,0}, \mathbf{v}_{c,1} + \Delta\mathbf{p}_{c,\text{upper}}, \mathbf{v}_{c,2} + \Delta\mathbf{p}_{c,\text{middle}}, \mathbf{v}_{c,3} + \Delta\mathbf{p}_{c,\text{lower}}], \mathcal{P}_{\text{suf}}(c)), \\
c &\in \{1, \dots, K\}.
\end{aligned} \tag{2}$$

$$\ell_{i,c}^{(1)} = \gamma \widehat{\mathbf{z}}_i^\top \widehat{\mathbf{t}}_c, \quad \mathcal{L}_{\text{S1A}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\ell_{i,y_i}^{(1)})}{\sum_{c=1}^K \exp(\ell_{i,c}^{(1)})}. \tag{3}$$

$$\begin{aligned}
\bar{\nu}_{c,r} &= \min(\mathbf{1}, \max(\mathbf{0}, \nu_{c,r})), & \mathbf{d}_{c,r}^+ &= \max(\mathbf{0}, \mathbf{d}_{c,r}), \\
\mathbf{u}_{c,r}^+ &= \max(\mathbf{0}, \mathbf{u}_{c,r}), \\
\mathbf{g}_{c,r} &= \text{Concat}(\mathbf{m}_{c,r} \odot \bar{\nu}_{c,r}, \log(\mathbf{1} + \mathbf{d}_{c,r}^+) \odot \bar{\nu}_{c,r}, \bar{\nu}_{c,r}, \mathbf{u}_{c,r}^+ \odot \bar{\nu}_{c,r}).
\end{aligned} \tag{4}$$

$$\Delta\mathbf{p}_{c,r} = \text{sigmoid}(a_r^{\text{prompt}}) q_{c,r} \tanh(f_r^{\text{prompt}}(\mathbf{g}_{c,r})), \quad r \in \{\text{upper}, \text{middle}, \text{lower}\}. \tag{5}$$

$$\begin{aligned}
\mathcal{L}_{\text{shift}} &= \frac{1}{KRD} \sum_{c=1}^K \sum_{r=1}^R \|\Delta\mathbf{p}_{c,r}\|_2^2, \\
\mathcal{L}_{\text{sem}} &= \max\left\{0, 1 - \frac{1}{K} \sum_{c=1}^K \cos(\widehat{\mathbf{t}}_c, \widehat{\mathbf{t}}_c^{\text{base}})\right\}, \\
\mathcal{L}_{\text{S1B}} &= \mathcal{L}_{\text{CE}}^{(1)} + \lambda_{\text{shift}} \mathcal{L}_{\text{shift}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem}}.
\end{aligned} \tag{6}$$

$$\begin{aligned}
s_{i,c}^{\text{cls}} &= (\mathbf{W}_{\text{cls}} \widehat{\mathbf{z}}_i^g + \mathbf{b}_{\text{cls}})_c, \\
s_{i,c}^{\text{global}} &= (\widehat{\mathbf{z}}_i^g)^\top \widehat{\mathbf{t}}_c, \\
s_{i,c}^{\text{local}} &= \sum_{r=1}^R \eta_{i,r} \widehat{\mathbf{h}}_{i,r}^\top \widehat{\mathbf{t}}_c, & \sum_{r=1}^R \eta_{i,r} &= 1.
\end{aligned} \tag{7}$$

$$\widetilde{s}_{i,c}^b = \frac{s_{i,c}^b}{\tau_b}, \quad \tau_b = \tau_{\min} + (\tau_{\max} - \tau_{\min}) \text{sigmoid}(\theta_b),$$

$$\begin{aligned}
\mathbf{w}_i &= \begin{cases} \boldsymbol{\pi}, & \text{fixed,} \\ \text{softmax}(\mathbf{a}), & \text{simplex,} \\ \text{softmax}(g(\widehat{\mathbf{z}}_i^g)), & \text{sample_dependent,} \end{cases} \\
w_{i,b} &\geq 0, & \sum_{b \in \mathcal{B}} w_{i,b} &= 1,
\end{aligned} \tag{8}$$

$$\begin{aligned}
s_{i,c} &= \sum_{b \in \mathcal{B}} w_{i,b} \widetilde{s}_{i,c}^b, \\
s_{i,c}^{\text{align}} &= \frac{1}{2} (\widetilde{s}_{i,c}^{\text{global}} + \widetilde{s}_{i,c}^{\text{local}}).
\end{aligned}$$

$$A_{i,r,p}^{\text{free}} = \text{softmax}_p \left(\frac{\mathbf{q}_r^\top \mathbf{W}_k \mathbf{z}_{i,p}}{\sqrt{D}} \right). \quad (9)$$

$$G_{i,r,j,p} = \exp \left[-\frac{\|\ell_{i,r,j} - \mathbf{c}_p\|_2^2}{2 \max\{10^{-8}, \sigma^2 (1 + u_{i,r,j}^+)^2\}} \right],$$

$$\tilde{A}_{i,r,p}^{\text{anat}} = \sum_j m_{i,r,j} \omega_{i,r,j} G_{i,r,j,p}, \quad (10)$$

$$A_{i,r,p}^{\text{anat}} = \frac{\tilde{A}_{i,r,p}^{\text{anat}}}{\max(\varepsilon_a, \sum_{p'} \tilde{A}_{i,r,p'}^{\text{anat}})}.$$

$$A_{i,r,p}^{\text{hyb}} = \rho_{i,r} A_{i,r,p}^{\text{anat}} + (1 - \rho_{i,r}) A_{i,r,p}^{\text{free}},$$

$$\mathbf{z}_{i,r}^r = \sum_p A_{i,r,p}^{\text{hyb}} \mathbf{z}_{i,p}. \quad (11)$$

$$\mathcal{L}_{\text{route}} = -\frac{\sum_{i,r} \rho_{i,r} \log \left[\max(\varepsilon_o, \sum_p A_{i,r,p}^{\text{free}} \text{sg}(A_{i,r,p}^{\text{anat}})) \right]}{\max(1, \sum_{i,r} \rho_{i,r})}, \quad (12)$$

$$\mathbf{e}_{i,r}^{\text{geo}} = \text{Concat}(\mathbf{x}_{i,r}^{\text{geo}} \odot \mathbf{m}_{i,r}^{\text{geo}}, \mathbf{m}_{i,r}^{\text{geo}}, \mathbf{u}_{i,r}^{\text{geo}}),$$

$$\mathbf{h}_{i,r} = \text{LN}_r[\mathbf{z}_{i,r}^r + \text{sigmoid}(a_r^{\text{vis}}) \rho_{i,r} \tanh(f_r^{\text{vis}}(\mathbf{e}_{i,r}^{\text{geo}}))]. \quad (13)$$

$$\eta_{i,r} = \text{softmax}_r(g_\phi(\text{Concat}(\mathbf{h}_{i,r}, \rho_{i,r}))), \quad s_{i,c}^{\text{local}} = \sum_{r=1}^R \eta_{i,r} \hat{\mathbf{h}}_{i,r}^\top \hat{\mathbf{t}}_c. \quad (14)$$

$$D_{\text{JSD},i} = H_\varepsilon(\bar{\mathbf{p}}_i) - \sum_{r=1}^R \bar{\omega}_{i,r} H_\varepsilon(\mathbf{p}_{i,r}), \quad H_\varepsilon(\mathbf{p}) = -\sum_{c=1}^K p_c \log(\max\{10^{-8}, p_c\}),$$

$$v_i^{\text{JSD}} = \mathbb{K} \left[\sum_r \rho_{i,r} \geq q_{\min} \right] \mathbb{K} \left[\sum_r \mathbb{K}(\rho_{i,r} > 0) \geq R_{\min} \right],$$

$$d_i^{\text{region}} = v_i^{\text{JSD}} \max(0, D_{\text{JSD},i}). \quad (15)$$

$$\mathbf{c}_i = \text{Concat}(\text{sg}(\hat{\mathbf{z}}_i^g), \boldsymbol{\rho}_i, \rho_i^{\min}, \rho_i^{\text{mean}}, \rho_i^{\text{valid}}, q_i^{\text{pose}}, q_i^{\text{crop}}) \in \mathbb{R}^{520}. \quad (16)$$

$$\tilde{r}_i = h_\psi(\mathbf{c}_i),$$

$$r_i = R_{\max} \tanh\left(\frac{\tilde{r}_i}{R_{\max}}\right),$$

$$S_i = \min\{S_{\max}, K + \text{softplus}(r_i)\},$$

$$\mathbf{p}_i = \text{softmax}(\mathbf{s}_i), \quad (17)$$

$$\alpha_{i,c} = 1 + \text{sg}(p_{i,c})(S_i - K),$$

$$\hat{p}_{i,c}^{\text{Dir}} = \frac{\alpha_{i,c}}{S_i},$$

$$u_i = \frac{K}{S_i}.$$

$$\mathcal{L}_{\text{cls}} = -\frac{1}{B} \sum_{i=1}^B \log \text{softmax}(\mathbf{s}_i)_{y_i}, \quad \mathcal{L}_{\text{align}} = -\frac{1}{B} \sum_{i=1}^B \log \text{softmax}(\mathbf{s}_i^{\text{align}})_{y_i}. \quad (18)$$

$$t_i = \mathbb{1} \left[\arg \max_c \text{sg}(s_{i,c}) = y_i \right], \quad \mathcal{L}_{\text{rel}} = -\frac{1}{B} \sum_{i=1}^B [t_i \log \text{sigmoid}(r_i) + (1 - t_i) \log(1 - \text{sigmoid}(r_i))]. \quad (19)$$

$$\mathcal{L}_{\text{rank}} = \frac{1}{B} \sum_{i=1}^B \max(0, m + r_i^{\text{corr}} - r_i^{\text{clean}}), \quad (20)$$

$$\mathcal{L}_{\text{S2}} = \mathcal{L}_{\text{cls}} + \beta_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{rel}} a_t \mathcal{L}_{\text{rel}} + \lambda_{\text{gate}} \mathcal{L}_{\text{gate}} + \lambda_{\tau} \mathcal{L}_{\text{temp}} + \lambda_{\text{route}} \mathcal{L}_{\text{route}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}. \quad (21)$$

$$\mathcal{L}_{\text{gate}} = D_{\text{KL}}(\bar{\mathbf{w}} \parallel \boldsymbol{\pi}), \quad \mathcal{L}_{\text{temp}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \left(\log \tau_b - \log \tau_b^{(0)} \right)^2, \quad (22)$$

$$\tilde{\boldsymbol{\alpha}}_i = \mathbf{y}_i + (\mathbf{1} - \mathbf{y}_i) \odot \boldsymbol{\alpha}_i, \quad \mathcal{L}_{\text{EDL}} = \frac{1}{B} \sum_{i=1}^B \left[\sum_{c=1}^K y_{i,c} (\psi(S_i) - \psi(\alpha_{i,c})) + a_t D_{\text{KL}}(\text{Dir}(\tilde{\boldsymbol{\alpha}}_i) \parallel \text{Dir}(\mathbf{1})) \right]. \quad (23)$$

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\hat{y}_i = y_i), \quad \text{BalancedAcc} = \frac{1}{K} \sum_{c=1}^K \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad (24)$$

$$\text{MacroF1} = \frac{1}{K} \sum_{c=1}^K \frac{2P_c R_c}{P_c + R_c}, \quad P_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad R_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}. \quad (25)$$

$$\text{ECE} = \sum_{m=1}^M \frac{|\mathcal{B}_m|}{N} |\text{acc}(\mathcal{B}_m) - \text{conf}(\mathcal{B}_m)|. \quad (26)$$

$$\kappa_j = \frac{j}{N}, \quad R_j = \frac{1}{j} \sum_{h=1}^j e_{(h)}, \quad \text{AURC}_{\text{code}} = \sum_{j=1}^{N-1} \frac{R_j + R_{j+1}}{2} (\kappa_{j+1} - \kappa_j). \quad (26)$$